

Deepfake Audio Detection with Statistical Classifiers

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Abstract—AI-driven voice synthesis have increased due to the rapid advancement in artificial intelligence (AI). Modern generative models can now produce speech that is highly realistic and often indistinguishable from genuine human audio and can be weaponised to spread false information or used to impersonate an individual, which makes deepfake audio detection crucial and of utmost importance. Statistical models are still useful because of their effectiveness, interpretability, and robustness for small to medium-sized datasets, even though Convolutional Neural Networks (CNNs) have shown great performance in audio classification tasks. In this work, we have trained and evaluated Support Vector Machine (SVM) and Logistic Regression (LR) models on the Fake-or-Real dataset. We combined these two statistical models into an ensemble framework and analyzed the performance against the singular models, and found that LR performs better in each scenario. LR is compared against a CNN baseline trained on the same dataset using well-established hyperparameters drawn from related research. Experimental results shows that LR achieves comparable accuracy and consistent performance under limited-data conditions, to CNN model, highlighting the effectiveness of lightweight statistical ensembles for AI audio deepfake detection.

Index Terms—Deepfake Audio Detection, Logistic Regression, Support Vector Machine

I. INTRODUCTION

The rapid advancement of artificial intelligence (AI) and generative models has led to a surge in synthetic media creation, particularly in the form of audio deepfakes. Modern speech synthesis and voice cloning technologies are capable of producing highly realistic human voices that are nearly indistinguishable from authentic recordings. While such technologies do offer great benefits in entertainment, accessibility, and virtual assistance, they also pose severe security, privacy, and misinformation risks. Malicious use of fake audio can lead to voice impersonation, financial fraud, and manipulation of public opinion, thereby emphasizing the urgent need for reliable AI audio detection systems.

The detection of AI-generated audio is a difficult task as human speech varies greatly, with different speakers, accents, and emotions. The AI-generated audio more often replicates human-like speeches with little distortion in the spectral or temporal properties. Due to its ability to acquire hierarchical and discriminative representations from raw or preprocessed information, deep learning-based methods, especially Convolutional Neural Networks (CNNs), have shown strong performance in large-scale audio classification challenges [1]. CNNs

often need large amounts of training data, a lot of computation, and careful architecture design, which may limit their practical use in low-data settings. On the other hand, traditional statistical learning models like Support Vector Machines (SVM) and Logistic Regression (LR) are still very applicable in fields where computational speed is crucial, interpretability is sought, and data is limited [2].

Logistic Regression is a linear probabilistic model that estimates the likelihood of a sample belonging to a particular class, making it well-suited for understanding how specific audio features contribute to classification outcomes. In contrast, the Support Vector Machine constructs a non-linear decision boundary in higher-dimensional space, often achieving better separation between classes in complex datasets. However, SVMs are also known to be more sensitive to the feature distribution and hyperparameter selection, which can result in model bias or instability if not carefully optimized [3].

In this work, we employ the Fake or Real (*For*) dataset to compare the performance of statistical models such as SVM and LR for fake audio detection. The focus of this work is to analyse different statistical model with different hyperparameters, examining their behavior while assessing the classification accuracy. Additionally, we explore an ensemble-based approach that combines the strengths of both SVM and Logistic Regression. By assigning differential weights to each model's predictions, we evaluate whether a weighted voting scheme can enhance overall detection performance compared to using either model individually. This comparison provides insights into the foundational differences between linear and non-linear statistical models in audio authenticity detection, setting the groundwork for deeper comparisons with neural network-based methods.

II. RELATED WORKS

Many of the pattern classification approaches have been employed to detect fake speech, including logistic regression (LR) [4], [5], random forest (RF) [6], and the support vector machine (SVM) [7].

The most used statistical classifiers in earlier work for audio detection is SVM due to its excellent classification capabilities. Alegre et al. [8] suggest that SVM classifiers are inherently robust to artificial signal spoofing attacks. However, it is very difficult to know the exact nature of fake attacks in practical

TABLE I: Initial Training Accuracy of Statistical Model

Metric	RF	SVM	LR
Accuracy (%)	45.13	63.24	63.69

scenarios. Therefore, Alegre et al. [9] and Villalba et al. [10] propose a one-class SVM classifier only trained using genuine utterances to classify real and fake voices, which generalizes well to unknown fake attacks.

The latest fake audio detection systems are mostly based on deep learning methods, which significantly outperform the SVM classifiers due to their powerful modelling capabilities [11]. The model architectures of classification are generally based on convolutional neural network (CNN), deep residual network (ResNet), modified ResNet (Res2Net), graph neural network (GNN), and Transformer.

CNNs are good at capturing spatially-local correlation, and CNN based classifiers have achieved promising performance [12], such as light CNN (LCNN) [13] consisting of convolutional and max-pooling layers with Max-FeatureMap (MFM) activation.

Some researchers attempt the CNN based models to end-to-end fake audio detection. Muckenhirn et al. [14] employed a simple CNN-based end-to-end approach to detection spoofed attacks. The proposed model consists of a single convolution layer and a multilayer perceptron (MLP) layer, which performs well for VC and TTS attacks.

In [15], the authors achieved an Equal Error Rate (EER) score of 0.03 on the LA dataset of ASVspoofing 2019 challenge, which is very competitive with state-of-the-art systems, by establishing the ensemble of selective spectrograms and models.

III. METHODOLOGY

We initially experimented with three classifiers—Random Forest (RF), Logistic Regression (LR), and Support Vector Machine (SVM)—using audio features extracted from the Fake or Real (FoR) dataset. Preliminary experiments revealed that the Random Forest model does not perform well with poor training accuracy as shown in Table 1. We excluded it from further experimentation and focused on LR and SVM, both of which demonstrated stable performance on the extracted features. We then performed systematic hyperparameter tuning for LR and SVM to obtain their optimal configurations, after which both models achieved comparable training accuracy. To further explore whether their complementary strengths could enhance detection reliability, we also constructed a weighted ensemble combining LR and SVM predictions. While the ensemble did not ultimately surpass the individual models in performance, it served as an important component for understanding in the evaluation pipeline.

A. Fake or Real Dataset

The Fake or Real (*FoR*) dataset is a publicly available dataset benchmark, aggregated from latest Text-To-Speech solutions as well as a variety of real human speech, for

evaluating fake audio detection systems. It contains an estimate of nearly 14,000 audio clips, equally divided between human speech and AI-generated voices. The dataset is structured into training, validation, and testing partitions, for each of its subset.

For training our SVM and LR models, we utilized the *for-2second* subset, which consists of uniformly trimmed 2-second audio samples. For inference and robustness testing, we employed the *for-rerecorded* subset—a re-recorded variant of the original clips designed to simulate a scenario where an attacker sends an utterance through a voice channel. This setup allows us to evaluate model performance under more realistic attack conditions.

B. Feature Extraction

For each audio file in the FoR dataset, we extract a total of 84 features, which are then standardized using a *StandardScaler* before being fed into the classification models. The feature set is designed to capture both spectral and temporal characteristics of speech, which helps in differentiating natural human audio from AI-generated ones.

Mel-Frequency Cepstral Coefficients (MFCCs) is the core of our feature representation. MFCCs represent the short-term power spectrum of sound, closely modeling human audio perception and helpful in effectively distinguishing tonal characteristics between real and fake audio. We compute thirteen base MFCCs along with their first-order derivatives (Δ MFCCs), allowing the model to capture not only the static frequency content but also its dynamic nature. **Chroma features** encode the distribution of energy across the twelve pitch classes of the chromatic scale, providing harmonic and tonal structure. In addition to these, we incorporate four spectrum- and energy-based descriptors: **spectral centroid**, which indicates the “brightness” of the signal; **spectral bandwidth**, which measures the spread of the frequency spectrum; **zero-crossing rate (ZCR)**, which reflects the presence of high-frequency components and abrupt transitions; and **root mean square (RMS) energy**, representing the overall loudness of the audio. These features help capture characteristics commonly associated with synthesis artifacts such as abrupt transitions, unnatural sharpness, or spectral smoothness.

Together, the MFCCs, Δ MFCCs, chroma vectors, and four spectral features form 42 primary frame-level descriptors. For each descriptor, we compute the mean and standard deviation across all frames in the audio clip, which results in an 84-dimensional feature vector per sample. Figure 1 shows the MFCC-spectrograms of a human-generated audio and that of a AI-generated audio.

C. Dimensionality Reduction

Chroma features captures the musical pitch and harmonic structure rather than the speech articulation or spectral characteristics. Because the *FoR* dataset consists exclusively

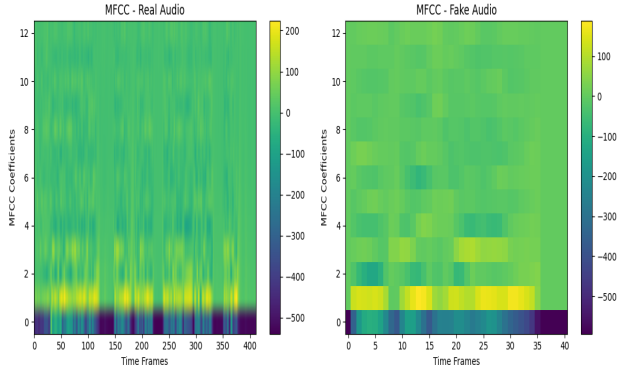


Fig. 1: MFCC spectrograms representations for human-generated audio and AI-generated Audio

TABLE II: Best Performing Hyperparameters

SVM			LR			
C	γ	Kernel	C	Penalty	Solver	Max Iteration
10	0.01	RBF	0.1	l1	liblinear	500

of speech samples with minimal tonal variation, we decided to remove the 12 chroma features and their corresponding statistical aggregations (mean and standard deviation), reducing the total feature count from 84 to 60. However, training the SVM on this reduced feature set resulted in a noticeable 7.5% drop in training accuracy, which shows that the model was sensitive to the removed feature space despite its limited semantic relevance to speech.

D. Hyper Parameters Tuning

To ensure optimal model performance, we performed hyperparameter tuning using GridSearchCV with a five-fold cross-validation ($cv = 5$) setup and F1-score as the evaluation metric. The best-performing hyperparameters obtained is as shown in Table 2.

The same hyperparameters are used for respective models with reduced parameters covered in Section 4.

E. Ensemble Learning

After training and validating both the models, we implemented an ensemble strategy using the trained SVM and Logistic Regression models, to check whether integrating multiple statistical learners can be more accurate than a single one. Each model contributes a weighted confidence score to the final decision. Several weight configuration were evaluated to identify the most effective balance between the two models.

IV. RESULTS AND COMPARISON

This section presents the experimental results obtained from comparison with our statistical models and the ensemble framework. We evaluate each model on both the original FoR-2second dataset and the FoR-rerecorded dataset to assess generalisation under real-world channel distortion conditions. Performance is measured using standard metrics such as

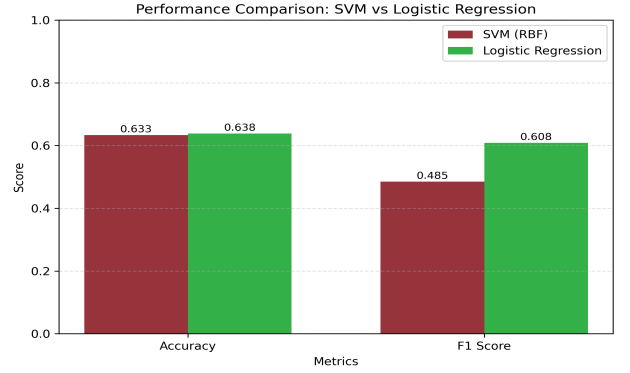


Fig. 2: Performance comparison of SVM and LR model during training

accuracy, precision, recall, and F1-score, with analysis of class-wise behavior to highlight model tendencies—including potential biases toward human or AI-generated speech. The results allow us to systematically compare linear, non-linear, and ensemble approaches, and to examine how lightweight statistical models perform relative to a deep learning baseline.

A. Comparing SVM and LR Pre-training

Figure 2 presents the training accuracy and F1 scores of both the SVM and Logistic Regression models. While the two models achieve almost identical accuracy, LR demonstrates a substantially higher F1 score—approximately 25% greater than that of SVM. This improvement indicates that LR maintains a better balance between precision and recall across both classes. In particular, LR is more effective at identifying the minority or harder-to-classify class, leading to fewer false positives and false negatives. Consequently, its higher F1 score reflects more reliable and consistent classification behavior, highlighting LR’s advantage in tasks where class imbalance and misclassification costs play an important role.

B. Inference Accuracy of SVM and LR

We first evaluated the trained SVM and LR models on a small, self-made audio dataset, where the SVM model achieved an accuracy of 66.67%, whereas the LR model correctly classified 100% of the samples, indicating its stronger generalisation to real-world conditions. To conduct a more robust and controlled evaluation, we then tested all four models—**SVM**, **SVM-Reduced**, **LR**, and **LR-Reduced**—on the *for-2rerec* split of the FoR dataset, evaluating the real and fake audio groups separately. Figure 3 illustrates the inference accuracy for each model on both classes.

SVM-Reduced achieved the highest accuracy, with SVM performing well on real audio but its performance deteriorates sharply for the fake audio classification, a 67% drop from its highest achieved accuracy. The best-performing SVM-Reduced model on real samples becomes the worst-performing on fake samples, which shows a strong *human bias* in SVM predictions. On the other hand, LR having lower accuracy than

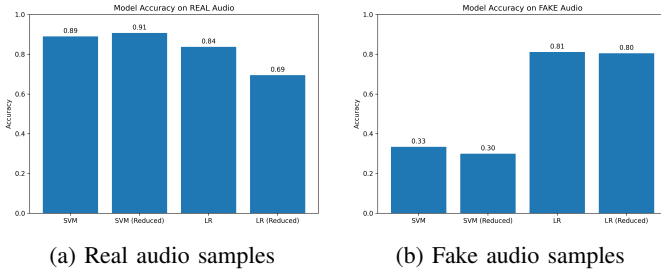


Fig. 3: Inference accuracy of SVM, SVM-Reduced, LR, and LR-Reduced on real and fake audio samples from the FoR rerecorded dataset.

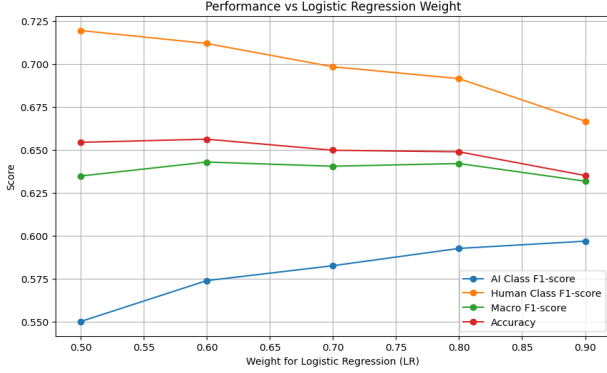


Fig. 4: Ensemble Analysis with varying LR_weights

that of SVM in real audio classification maintains its accuracy for the fake audio classification, showing robustness and a balanced classification behaviour. **LR seems to be the best-performing model and optimal choice among the 4 models trained.**

C. Analysis of Ensemble Framework

A weighted ensemble of SVM and LR models were experimented upon to find whether the combined SVM and LR model can overcome the bias-ness of SVM, and also achieving a good and balanced performance in both real and fake audio groups. LR_weights were varied in range of 0.5 to 1, and corresponding SVM_weights were as follows:

$$\text{SVM}_{\text{weights}} = 1 - \text{LR}_{\text{weights}}.$$

For each weight configuration, we computed the AI-class accuracy, human-class accuracy, macro F1-score, and overall accuracy, as summarized in Figure 4.

Based on these results, we selected two configurations—0.9 LR weight and 0.7 LR weight—for deeper analysis, namely LORIS9 and LORIS7 respectively.

Figure 5 shows the inference accuracy of LORIS7 and LORIS9 for real and fake audio classification. Both ensembles demonstrated improved performance over standalone LR, 5% and 1% increase for LORIS7 and LORIS9 respectively in classifying real audio samples, although they did not reach the performance of SVM on this class. However, for the fake audio classification, the performance drop was 9% and

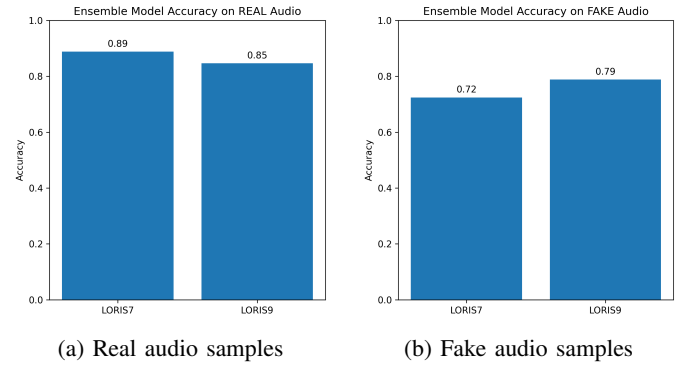


Fig. 5: Inference accuracy of LORIS7 and LORIS9 on real and fake audio samples from the FoR rerecorded dataset.

2% in accuracy, for LORIS7 and LORIS9 respectively when compared to standalone LR.

Overall, the ensemble did not provide a sufficiently strong improvement to justify its use. The gain in real-audio accuracy was marginal, while the drop in fake-audio accuracy made the combined model less reliable. Consequently, **LR alone remained the most balanced and robust choice for the final model.**

V. CONCLUSION

We have conducted a comprehensive analysis of statistical machine learning models for fake audio detection using the Fake-or-Real (FoR) dataset. By extracting 84 domain features and evaluating multiple classifiers, we identified that Logistic Regression (LR) provide as the most robust and reliable model. While both SVM and LR have shown similar performance during training, the inference experiments reveals that SVM has a significant bias towards real audio samples and a poor generalization for fake audio detection.

We also implemented a weighted ensemble of SVM and LR to determine if combining models could improve performance and remove bias-ness of SVM. Although the ensemble improved the real-audio classification marginally, it still underperformed LR on fake audio. The marginally improvement was not sufficient enough compared to LR. Logistic Regression offers the best balance of precision, recall, and generalization across both real and fake audio classes. This study highlights the importance of analyzing class-specific behavior and not relying solely on aggregate accuracy. Future work can explore deep learning models to further strengthen fake audio detection systems.

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